ATOMOD

User guide

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Installation

Before you begin, check the ATOMOD systems requirements, to avoid any issues during the process. Ensure you have the following:

- PYTHON:
  - Requires Python 3.12 or an earlier version (e.g., 3.11, 3.10) to avoid compilation errors.
- DISK SPACE:
  - At least 7 GB of free space on your partition.
- SYSTEM SOFTWARE (Required):
  - `git` (to clone the ATOMOD repository)
  - `python3-venv` (to isolate Python packages)
  - `python3-pip` (to install dependencies)
  - `wget` (to download installer `install.py`)

  On Ubuntu / Xubuntu / Debian, you can install everything using:
  `sudo apt update && sudo apt install git python3-pip python3-venv wget`

Windows

Installation process on windows:

- 1 - Open a PowerShell (Fig.1(1-2-3))
- 2 - Navigate to the directory of your choice (e.g., `cd ~/Desktop` or `cd ~/src`)(Fig.1(4)).

  ```
  cd C:\Users\herve\Desktop\
  ```

- 3 - Download the installation script `install.py`.

  ```
  wget https://github.com/hbulou/ATOMOD/releases/download/v0.1/install.py
  ```

- 4 - Run the installation (Fig.1(5)). NOTE: You must have at least 7 GB of available disk space.

  ```
  python.exe install.py
  ```

- 5 - Once the installation is complete, go to the ATOMOD directory (Fig.1(6)).

  ```
  cd ATOMOD
  ```

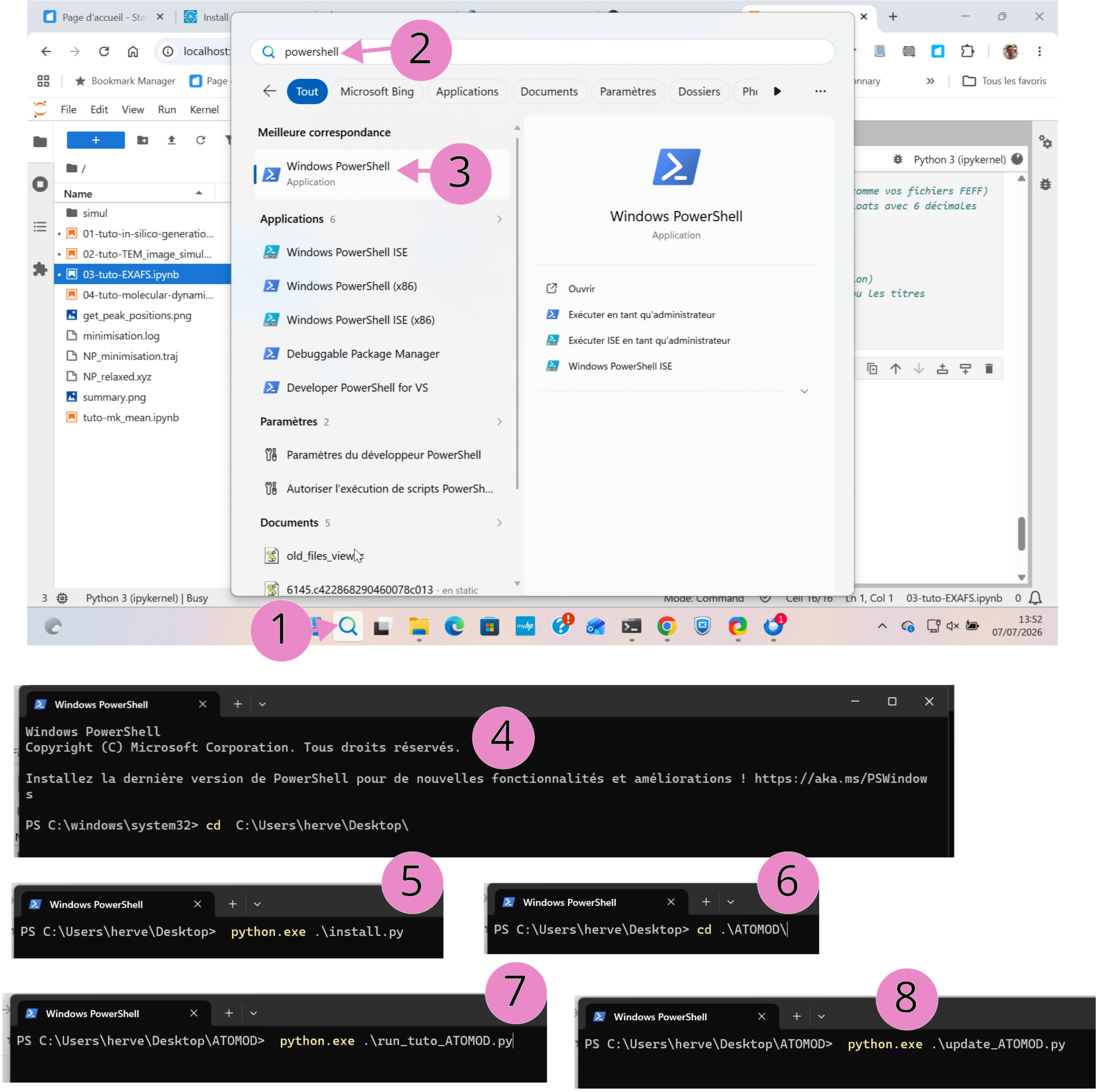

Figure 1: Screenshots of the various stages of installing ATOMOD on Windows.

Linux

Installation process on linux:

- 1 - Open an X terminal
- 2 - Navigate to the directory of your choice (e.g., `cd ~/Desktop` or `cd ~/src`)(Fig.2(a)).

  `cd Desktop`
- 3 - Download the installation script `install.py` (Fig.2(b)).

  `wget https://github.com/hbulou/ATOMOD/releases/download/v0.1/install.py`
- 4 - Run the installation (Fig.2(c)). NOTE: You must have at least 7 GB of available disk space.

  `python3 install.py`

- 5 - Once the installation is complete, go to the ATOMOD directory (Fig.2(d)).

  `cd ATOMOD`

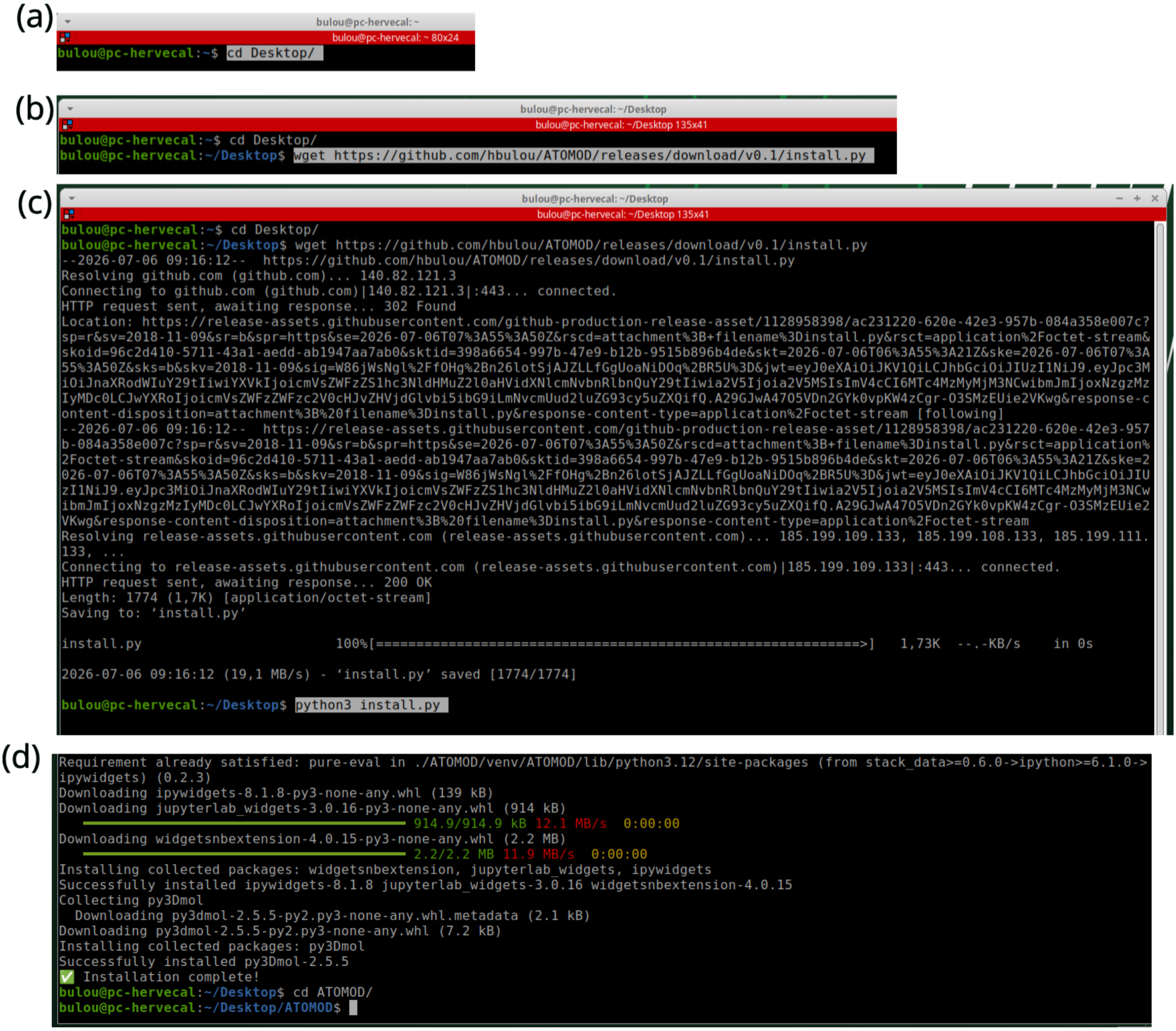

Figure 2: Screenshots of the various stages of installing ATOMOD on Linux.

ATOMOD Update

To update ATOMOS, simply run the `update_ATOMOD.py` script (Fig.1(8)):
`python3 update_ATOMOD.py`

Tutorials

Four tutorials are available in the `./doc/tutorials` directory:

- `01-tuto-in-silico-generation.ipynb`, describing the procedure for creating a nanoparticle *in silico*,
- `02-tuto-TEM_image_simulation.ipynb`, describing the procedure for simulating a TEM image,
- `03-tuto-EXAFS.ipynb`, describing the procedure for simulating the EXAFS spectra of a nanoparticle,

- `04-tuto-molecular-dynamics.ipynb`, describing the procedure for performing classical molecular dynamics (MACE force fields).

To run the tutorials, the simplest method is to use the `run_tuto_ATOMOD.py` file located in the `ATOMOD` directory ((Fig.1(7)):

```
python3 run_tuto_ATOMOD.py
```

Note regarding `03-tuto-EXAFS.ipynb`: the tutorial was developed for use with `FEFF` installed outside the `ATOMOD` distribution; therefore, this tutorial will not work if you do not have `FEFF` installed on your computer.